Characteristics, Caustics, and the Selection of Shocks

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A first-order PDE often becomes an ordinary differential equation when followed along the right curves. For linear transport these curves simply carry an initial profile; for nonlinear conservation laws their speeds depend on the transported value itself. When faster characteristics catch slower ones, the classical solution ceases to exist. The weak conservation law still makes sense, but it admits too many solutions until an entropy condition selects the physically relevant one.[1]

§1.1 The chain rule chooses the path

Consider

$$u_t + a(x,t)u_x = b(x,t), \qquad u(x,0) = f(x),$$

where the coefficients are sufficiently smooth to give unique characteristic curves. If $x(t)$ solves

$$x'(t) = a(x(t),t),$$

then the chain rule gives

$$\frac{d}{dt}u(x(t),t) = u_t + a(x(t),t)u_x = b(x(t),t).$$

The PDE is therefore equivalent, along each characteristic, to a pair of ODEs:

$$x'(t) = a(x(t),t), \qquad \frac{d}{dt}u(x(t),t) = b(x(t),t).$$

For constant $a = c$ and no forcing, $x(t) = \xi + ct$ and

$$\boxed{u(x,t) = f(x-ct).}$$

This is a translating profile: neither its height nor its shape changes with time.

As a small example, take $u_t + cu_x = x + t$. Along the characteristic through (x,t), the earlier position is $x - c(t-s)$. Thus

$$\begin{aligned} u(x,t) &= f(x-ct) + \int_0^t (x - c(t-s) + s)\, ds \\ &= \boxed{f(x-ct) + xt + \frac{1-c}{2}t^2}. \end{aligned}$$

The force accumulates along a moving path, rather than at a fixed spatial point.

[1] G. Strang, MIT 18.086 (*Characteristics, Shocks, and Fans*); MIT 18.306, lectures on shock and entropy solutions.

§1.2 Characteristic data require transversality

Suppose initial values are prescribed on a parametrized curve $s \mapsto (x_0(s), t_0(s))$. The directional derivative specified by the PDE is along $(a, 1)$ in the (x, t)-plane. To recover an entire neighborhood from the initial curve, that curve must cross the characteristics:

$$\det \begin{pmatrix} x_0'(s) & a(x_0(s), t_0(s)) \\ t_0'(s) & 1 \end{pmatrix} = x_0'(s) - a(x_0(s), t_0(s)) t_0'(s) \neq 0.$$

This is the noncharacteristic condition. It ensures the characteristic flow gives local coordinates built from the initial curve parameter and time.

For $u_t + u_x = 0$, prescribing data on $x - t = 0$ is not enough to determine the solution nearby: that line is itself a characteristic. Any compatible data must be constant along it, and the values on neighboring lines remain unspecified. The failure comes from geometry, not from an algebraic defect in the formula.

§1.3 Nonlinear characteristics can collide

The inviscid Burgers equation is

$$u_t + u u_x = 0, \qquad u(x, 0) = u_0(x).$$

Along a characteristic, the value u stays constant. If the characteristic begins at ξ, its speed is $u_0(\xi)$, so

$$\boxed{x = \xi + t\, u_0(\xi), \qquad u(x, t) = u_0(\xi).}$$

As long as the map $\xi \mapsto x$ remains invertible, this implicitly defines a classical solution.

Differentiate the characteristic relation with respect to ξ:

$$\frac{\partial x}{\partial \xi} = 1 + t\, u_0'(\xi).$$

The first crossing occurs when this Jacobian vanishes. Differentiating the solution itself gives

$$\boxed{u_x(x, t) = \frac{u_0'(\xi)}{1 + t\, u_0'(\xi)}.}$$

A negative initial slope therefore steepens: the denominator shrinks toward zero and the spatial derivative blows up while the solution values remain bounded.

Take the smooth initial profile $u_0(\xi) = -\tanh \xi$. Since

$$u_0'(\xi) = -\operatorname{sech}^2 \xi,$$

the first characteristic crossing occurs at

$$t_* = -\frac{1}{\min_\xi u_0'(\xi)} = 1.$$

Near $\xi = 0$, the characteristic map is

$$x = \xi - t \tanh \xi = (1 - t)\xi + \frac{t}{3}\xi^3 + O(\xi^5).$$

At $t = 1$ the linear term disappears, making the projection of the characteristic family singular. This is a caustic: the map from initial labels to physical positions loses invertibility. For $t > 1$, the formal characteristic solution becomes multivalued near $x = 0$.

§1.4 Conservation determines the speed of a jump

Burgers' equation is also the conservation law

$$\boxed{u_t + \partial_x\left(\frac{u^2}{2}\right) = 0.}$$

A discontinuity is not differentiable in the classical sense, but the integral conservation law survives.

For a general flux F, suppose u has a jump at $x = s(t)$, with left and right states u_L, u_R. Integrate the conservation law over a narrow interval straddling the moving jump. As the interval shrinks, the balance is

$$s'(t)(u_L - u_R) = F(u_L) - F(u_R).$$

Thus the Rankine–Hugoniot jump speed is

$$\boxed{s'(t) = \frac{F(u_L) - F(u_R)}{u_L - u_R}} \qquad (u_L \neq u_R).$$

For Burgers' flux $F(u) = u^2/2$, this becomes

$$\boxed{s' = \frac{u_L + u_R}{2}.}$$

Conservation fixes a discontinuity's speed, but it does not by itself say whether a proposed discontinuity is physically admissible.

§1.5 Shocks versus Rarefactions

Consider the step initial data

$$u(x,0) = \begin{cases} u_L, & x < 0, \\ u_R, & x > 0. \end{cases}$$

When $u_L > u_R$, left characteristics travel faster than right ones and converge. The entropy solution has a shock moving at $s = (u_L + u_R)/2$:

$$u(x,t) = \begin{cases} u_L, & x < st, \\ u_R, & x > st. \end{cases}$$

For the concrete data $u_L = 1$, $u_R = 0$, the shock speed is $1/2$. The characteristics move into it:

$$1 = F'(u_L) > \frac{1}{2} > F'(u_R) = 0.$$

This inequality is the Lax compressivity condition for a convex flux.

When $u_L < u_R$, characteristics separate, leaving a fan. For $u_L = 0, u_R = 1$, the entropy solution is

$$\boxed{u(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t, \end{cases} \qquad t > 0.}$$

Inside the fan, $u = x/t$ satisfies $u_t + uu_x = 0$ because

$$u_t = -\frac{x}{t^2}, \qquad uu_x = \frac{x}{t} \cdot \frac{1}{t} = \frac{x}{t^2}.$$

The profile is continuous across both fan edges and has the appropriate limiting initial data.

For the same increasing step, a discontinuity of speed $1/2$ also satisfies Rankine–Hugoniot, yet is not the entropy solution. The missing selection criterion is therefore substantive, not merely a preferred way of drawing characteristics.

§1.6 Entropy and Time Direction

For smooth Burgers solutions, every convex twice-differentiable function $\eta(u)$ satisfies another conservation law:

$$\partial_t \eta(u) + \partial_x q(u) = 0, \qquad q'(u) = \eta'(u)F'(u) = u\eta'(u).$$

For discontinuous solutions, the entropy condition replaces equality by

$$\boxed{\partial_t \eta(u) + \partial_x q(u) \le 0}$$

in the distributional sense, for every convex entropy pair.

Take $\eta(u) = u^2/2$, so $q(u) = u^3/3$. Across a Burgers shock of speed $s = (u_L + u_R)/2$, the coefficient of the delta distribution in $\eta_t + q_x$ is

$$[q] - s[\eta] = \frac{u_R^3 - u_L^3}{3} - \frac{u_L + u_R}{2}\frac{u_R^2 - u_L^2}{2} = \frac{(u_R - u_L)^3}{12}.$$

It is nonpositive exactly when $u_L \ge u_R$. The decreasing step creates an entropy-admissible shock; the increasing step would give positive entropy production in this convention and is replaced by a rarefaction fan.

This calculation makes an important distinction. Characteristics explain how smooth information travels, and their crossing predicts failure of a classical solution. Conservation fixes the possible jump speed, while entropy rules out the unphysical weak solutions. The full theory needs all three levels because each answers a different mathematical question.